

ASSIGNMENTS**Class-1****Vector Space and Subspace**

1. Define the vector space with example.
2. Define the sub space with example.
3. Write all the subspaces of $\mathbb{R}^3$.
4. Show that the set of non-singular 2×2 matrices is not a subspace.
5. Check whether the following subset of $\mathbb{R}^3$ is subspace or not

The plane of vector b with $b_1 = 1$

$$\text{i.e. } S = \left\{ \begin{bmatrix} 1 \\ b_2 \\ b_3 \end{bmatrix} / b_1 = 1, b_2, b_3 \in \mathbb{R} \right\}$$